

MUHAMMAD ADEEL SAJJAD

SYSTEMS, INFRASTRUCTURE & CYBER-PHYSICAL SECURITY ENGINEER

B.Sc. ELECTRICAL ENGINEERING — POWER

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1. PROFILE / SUMMARY

Systems & Infrastructure Engineer with a foundational degree in Electrical Engineering (Power Systems) and over 8 years of enterprise infrastructure experience. Rigorous training in electromagnetic field modeling, resonant power electronics, and high-voltage transmission, combined with deep operational expertise in deterministic kernel telemetry pipelines, bare-metal infrastructure automation, and industrial control systems (SCADA) network boundary defense.

2. RESEARCH INTERESTS

- Resilient Cyber-Physical Systems (CPS):** Analyzing failure modes & attack surfaces where unauthenticated industrial protocols (Modbus-TCP, DNP3) couple with physical power converter dynamics and protection relays.
- Smart Grid Security & Anomaly Detection:** Deterministic packet inspection, state-estimation defense against false data injection (FDI), and IEC 62443 / Purdue reference architecture enforcement.
- Resonant Inductive Coupling:** High-Q magnetic resonance link optimization, mutual inductance parameter extraction, and zero-voltage-switching (ZVS) power electronics.
- Distributed Testbed Infrastructure:** Hardware-in-the-loop (HIL) environments, kernel-level zero-drop ring buffering, and reproducible bare-metal infrastructure orchestration.

3. EDUCATION & ACADEMIC FOUNDATION

Bachelor of Science in Electrical Engineering — Power Systems

Graduated 2016

The Islamia University of Bahawalpur — Pakistan • Specialization: Power Systems & Electrical Infrastructure • Status: Degree Conferred (2016)

Capstone Thesis: "Resonant Inductive Wireless Power Transmission System" (Aug 2015 — Jun 2016, Grade A)

Modeled Maxwell's equations and coupled-mode theory for near-field electromagnetic energy transfer over variable air gaps. Designed, wound, and tuned helical magnetic induction coils. Developed primary and secondary resonant LC tanks to cancel reactive power and maximize power factor. Fabricated switching driver hardware and evaluated transmission efficiency curves, Q factors, and thermal behavior using DSO oscilloscopes and RF instruments. Authored thesis documentation followed by an examination-board defense.

4. PROFESSIONAL ENGINEERING EXPERIENCE

Senior IT & Infrastructure Systems Manager

Nov 2025 — Present

Urban 411 Design Products Trading · Systems & Infrastructure Operations — Dubai, UAE

Directing enterprise IT and cloud systems infrastructure with complete lifecycle management across multi-tier hosting, automated telemetry pipelines, perimeter hardening, and automated CI/CD deployments.

- Achieved 40%+ reduction in server response latency through reverse-proxy caching, database query optimization, and kernel parameter tuning.
- Enforced a 99.8% operational SLA across multi-tier infrastructure by eliminating single points of failure and automating recovery.
- Deployed centralized Wazuh SIEM telemetry pipeline ingesting host events and audit logs with automated threat detection rules.

Systems: Wazuh HIDS/SIEM/XDR Security Platform, Linux Distributions (Ubuntu Server, Debian, RHEL), Nginx & Apache Web Servers, GitHub Actions & Bash CI/CD Workflows, Certbot & Let's Encrypt Automated SSL/TLS, Multi-Tier Cloud Hosting (AWS EC2, Lightsail)

IT & Infrastructure Operations Manager

2020 — 2025

TECDEAL General Trading LLC · Infrastructure & Operations — Dubai, UAE

Led infrastructure architecture and systems operations across distributed commercial hubs, securing local network topologies, gateway firewalls, database clusters, and virtualization environments.

- Segmented distributed commercial hub networks using VLANs and gateway ACLs, completely isolating sensitive transactional servers.
- Optimized relational database query execution plans and index trees, eliminating peak-hour query bottlenecks.
- Implemented automated multi-region snapshot failover workflows ensuring rapid recovery in the event of local infrastructure disruption.

Systems: Gateway Firewalls & Multi-VLAN Subnetting, Server Virtualization Environments, Relational Database Clusters (MySQL / PostgreSQL), Multi-Region Cloud Snapshot Pipelines, Hardware Access Control & Bastion Proxies

Systems & Web Development Engineer

2016 — 2019

Techvision · Systems & Web Engineering — Bahawalpur, Pakistan

Managed production Linux LAMP/LEMP server environments, performing routine security audits, system patching, automated shell scripting, and local network maintenance.

- Maintained consistent server uptime through automated cron diagnostics, preemptive disk space monitoring, and offsite log archiving.
- Executed systematic security patching schedules across production Linux servers, eliminating exposed service vulnerabilities.
- Designed reliable shell-driven offsite database backup replication routines with automated integrity verification.

Systems: Linux LAMP & LEMP Production Stacks, Apache & Nginx Web Servers, MySQL Database Administration, Bash Shell & Cron Automation Daemons, Physical Routing & Managed Switch Hardware

5. SELECTED PROJECTS & CYBER-PHYSICAL SYSTEMS

[PROJECT / WPT-001] Resonant Inductive Wireless Power Transmission System Aug 2015 — Jun 2016 / CAPSTONE THESIS
Domain: Electrical Engineering / Electromagnetic Modeling • **Technologies:** Coupled-Mode Theory, LC Resonant Tanks, ZVS Inverter, Helical Coils
Modeled Maxwell's equations and coupled-mode theory for near-field electromagnetic energy transfer over variable air gaps (0–150 mm). Designed, wound, and tuned helical magnetic induction coils. Developed primary and secondary resonant LC tanks to cancel reactive power and maximize power factor. Fabricated switching driver/inverter hardware with gate-drive isolation. Evaluated transmission efficiency curves, Q factors, and thermal behavior using DSO oscilloscopes, RF power meters, and signal generators. Authored thesis documentation containing mathematical derivations, schematic designs, and empirical performance metrics, defended before the examination board.
Verification: Galvanic isolation metrology • Repository: GitHub / wireless-power-transmission-sim

[PROJECT / CPS-001] MICROGRID RESILIENT CONTROL 2025 / OPEN-SOURCE CPS REPOSITORY
Domain: Cyber-Physical Systems / Electrical Infrastructure / Control Security • **Technologies:** Inverter Droop Dynamics (P-f / Q-V), Distributed Consensus Graph Theory, Byzantine Resilient Filtering, Python CPS Dynamics Suite
Secondary frequency consensus restoration on islanded AC microgrids under False Data Injection (FDI) attacks. Features distributed cooperative control algorithms for inverter-based distributed energy resources (DERs), modeling physical frequency droop dynamics coupled with resilient state exchange across communication topologies. Incorporates detection and mitigation of adversarial data corruption targeting consensus communication links.
Verification: Bandit SAST security scanning • Repository: GitHub / microgrid-resilient-control

[PROJECT / CPS-002] BATTERY EKF STATE ESTIMATOR 2025 / OPEN-SOURCE CPS REPOSITORY
Domain: Cyber-Physical Systems / Battery Systems / Embedded Security • **Technologies:** Discrete Extended Kalman Filter (EKF), Equivalent Circuit Model (Thevenin ECM), CAN Bus (ISO 11898) Parser, Boundary Validation Rules
Discrete Extended Kalman Filter (EKF) for Li-ion State-of-Charge estimation combined with boundary-checked CAN-bus telemetry parsing and frame fuzzing. Implements an equivalent circuit model (Thevenin ECM) capturing battery polarization dynamics and nonlinear open-circuit voltage curves. Pairs filtering mathematics with defensive embedded software engineering, including frame validation, range boundaries, and fuzzing resistance.
Verification: Stochastic convergence verification across noisy drive cycles • Repository: GitHub / battery-ekf-state-estimator

[PROJECT / CPS-003] SCADA IDS TELEMETRY ENGINE 2025 / OPEN-SOURCE CPS REPOSITORY
Domain: Industrial Security / SCADA / OT Security • **Technologies:** Modbus TCP Protocol Parser, APDU Header Dissection, Zero-Trust Command Whitelisting, Physical Domain Rule Engine
Deep-packet APDU dissector for Modbus TCP/IEC SCADA protocols with zero-trust command whitelisting and substation physical anomaly detection. Inspects industrial network traffic at the application protocol data unit (APDU) layer, enforcing zero-trust command policies (e.g. blocking unauthorized coil writes or register overrides) and verifying telemetry against physical domain constraints.
Verification: Memory safety fuzzing on malformed MBAP headers and truncated APDU payloads • Repository: GitHub / scada-ids-telemetry-engine

[PROJECT / SEC-001] QUANTUM PQC TLS CANARY 2025 / OPEN-SOURCE SECURITY REPOSITORY
Domain: Post-Quantum Cryptography / CPS Security / Network Security • **Technologies:** NIST FIPS 203 ML-KEM-768, X25519 Hybrid Key Exchange, Cryptographic Key Derivation (HKDF), TLS Canary Instrumentation
NIST FIPS 203 ML-KEM-768 + X25519 hybrid key encapsulation canary protecting edge CPS telemetry against Harvest-Now-Decrypt-Later (HNDL) attacks. Explores post-quantum cryptographic primitives on constrained communication channels, measuring overhead, handshake latency, and TLS encapsulation behavior.
Verification: Downgrade attack detection test harness • Repository: GitHub / quantum-pqc-tls-canary

[PROJECT / SYS-004] Infrastructure Automation & CI/CD Pipelines 2025 / AUTOMATION WORKFLOWS
Domain: Enterprise Infrastructure / Distributed Systems • **Technologies:** GitHub Actions, Bash Shell Scripting, Python Automation, SSH Transport
Architected automated deployment workflows and infrastructure synchronization pipelines using Bash, Python, and GitHub Actions. Automated secure SSH/FTP delivery, environment health validation, configuration templating, and operational maintenance routines to guarantee reliable application delivery.
Verification: Encrypted SSH deployment keys with restricted commands • Repository: GitHub / Haidriyam

6. TECHNICAL NOTES & ENGINEERING MONOGRAPHS

- [TN-001]** "Resonant Inductive Coupling: Efficiency, Frequency and Air-Gap Analysis"
Category: Electromagnetics & Resonance
Rigorous mathematical derivation of coupled magnetic resonance in Series-Series (SS) inductive links. Quantifies mutual inductance decay over variable air gaps, establishes the critical coupling threshold separating single-mode and bifurcation regimes, and analyzes conductor AC resistance dissipation.
- [TN-002]** "Distributed Network Telemetry and Intrusion Detection"
Category: Telemetry & Distributed Systems
Architecture of deterministic telemetry ingestion pipelines for cyber-physical infrastructure; kernel socket buffer tuning (SO_RCVBUFFORCE), AF_PACKET MMAP ring buffers, and deterministic APDU regex tokenization under burst traffic.
- [TN-003]** "SCADA Network Security Considerations for Power Infrastructure"
Category: Industrial SCADA & CPS Security
Cyber-physical vulnerability and boundary inspection analysis of unauthenticated industrial fieldbus protocols (Modbus-TCP, DNP3, IEC 60870-5-104); prevention of false data injection (FDI) and MITM coil forced writes via IEC 62443 zone segmentation.

7. TECHNICAL COMPETENCIES

Power Systems & Hardware: High-Q Resonant Inductive Coupling, Series-Series Tank Compensation, zero-voltage-switching (ZVS) inverters, transmission fault studies, DSO oscilloscopes, precision LCR meters, and spectrum analyzers.
Infrastructure, Security & Networking: Linux kernel sockets (SO_RCVBUFFORCE), AF_PACKET MMAP ring buffers, SCADA Modbus-TCP / DNP3 deep packet inspection, IEC 62443 zone segmentation, Ansible automation, and WireGuard mesh.

8. VERIFIED CURRICULA & CREDENTIALS

Packt / Coursera: AWS Certified DevOps Engineer – Professional Specialization	Verified Curriculum	IBM: Introduction to Hardware and Operating Systems	Verified Curriculum
IBM: Introduction to Networking and Storage	Verified Curriculum	IBM: Introduction to Cybersecurity Essentials	Verified Curriculum
IBM: Cybersecurity Tools & Cyberattacks	Verified Curriculum	Northeastern University: Software Quality Control and Management Parts 1 & 2	Verified Curriculum
University of Leeds: Foundations of Software Testing and Validation	Verified Curriculum	Meta: Advanced Programming in Kotlin	Verified Curriculum
IBM: Introduction to Software, Programming, and Databases	Verified Curriculum	Skillshare / Coursera: Modern Frontend Web Development Specialization	Verified Curriculum

REFERENCES & INSTITUTIONAL VALIDATION:Academic supervisor recommendations, authenticated degree transcripts, and institutional validation records available immediately upon request during formal evaluation.